# impl Sample for ZExpFamily<2>

Michael Shoemate (@Shoeboxam)

This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of Sample for ZExpFamily<2> in mod.rs at commit f5bb719 (outdated[1]).

## 1 Hoare Triple

### Precondition

#### Compiler-verified

None

#### Caller-verified

self represents a valid probability distribution.

### Pseudocode

```
1 class ZExpFamily2:  # analogous to impl Sample for ZExpFamily<1> in Rust
2     def sample(self, shift):
3         return shift + sample_discrete_gaussian(self.scale)  #
```

### Postcondition

**Theorem 1.1.** Either returns Err(e) independently of the input shift, or Ok(shift + Z) where Z is a sample from the distribution defined by self.

*Proof.* By the precondition, since self represents a valid probability distribution, then by the definition of ZExpFamily, self.scale is non-negative.

Since the preconditions for are met (non-negative scale), then we claim the postcondition that the returned value follows a given distribution. This distribution matches that defined by self.

By the addition on line 3, the implementation satisfies the postcondition. □

[1] See new changes with git diff f5bb719..12d5646 rust/src/measurements/noise/mod.rs